# Week 5: Faster Sampling with DDIM

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| **Theme:** | 1000 sampling steps is too many. Cut it to 50 with no quality loss. |
| **Time Commitment:** | 8–10 hours (lighter recovery week after Week 4) |
| **Suggested Split:** | 3h reading, 5h coding/benchmarking, 1h writeup |

## Learning Objectives

By the end of this week, you should be able to:

- Articulate the difference between stochastic (DDPM) and deterministic (DDIM) sampling
- Implement DDIM sampling on top of your existing pre-trained DDPM
- Benchmark sampling speed vs. quality trade-offs systematically
- Understand the role of the $\eta$ (eta) parameter

## Primary Resources (Must-Do)

### [PAPER] DDIM Paper: "Denoising Diffusion Implicit Models" (Song et al., 2021)

**Type:** Research Paper | **Time:** 90 min | **Difficulty:** Advanced
**URL:** https://arxiv.org/abs/2010.02502

**Why this resource:** The original DDIM paper. Surprisingly accessible if you've internalized DDPM. The key insight: the same trained model can be sampled deterministically with far fewer steps.

**Focus on:** Sections 1–4. Skip the proofs in the appendix unless you're curious. The sampling equation is what you'll implement.

### [BLOG/DOCS] Lilian Weng: DDIM Section

**Type:** Blog Post | **Time:** 30 min | **Difficulty:** Intermediate-Advanced
**URL:** https://lilianweng.github.io/posts/2021-07-11-diffusion-models/#speed-up-diffusion-model-sampling

**Why this resource:** Concise companion to the paper. Focuses on the practical implementation differences between DDPM and DDIM sampling.

### [CODE] Hugging Face Diffusers: DDIMScheduler Source

**Type:** Code | **Time:** 30 min | **Difficulty:** Intermediate
**URL:** https://github.com/huggingface/diffusers/blob/main/src/diffusers/schedulers/scheduling_ddim.py

**Why this resource:** Production-grade reference implementation. Read the `step()` method carefully — compare it to your DDPM implementation.

**Focus on:** Lines around the variance calculation and the use of `eta`.

### [VIDEO] AI Coffee Break: "DDIM Explained"

**Type:** YouTube Video | **Time:** 15 min | **Difficulty:** Intermediate
**URL:** https://www.youtube.com/watch?v=344w5h24-h8

**Why this resource:** Quick, visual summary of why DDIM works. Watch this before diving into the paper for an overview.

### This Week's Deliverable

Add DDIM sampling to your existing DDPM model from Week 4. **No retraining needed** — this is the magic. Code must include:

- A `DDIMScheduler` class with configurable `eta` parameter (where $\eta = 0$ is fully deterministic, $\eta = 1$ recovers DDPM)
- A method to compute the appropriate timestep subset (evenly spaced from $[0, T]$)
- Side-by-side comparison: DDPM at 1000 steps vs. DDIM at 10, 25, 50, and 100 steps
- Wall-clock timing for each setting (use `time.time()` or `torch.cuda.Event`)
- A short markdown report (`ddim_report.md`) with sample grids and a timing table
- Bonus: an interpolation experiment — pick two random noise tensors and linearly interpolate between them, decoding each step. With deterministic sampling, you get smooth transitions.

### Key Concepts Checklist

You should be able to confidently answer all of the following:

- Why can DDIM use the same trained model as DDPM?
- What does the $\eta$ parameter control? What happens at $\eta = 0$ vs. $\eta = 1$?
- Why is interpolation between noise vectors only meaningful with deterministic sampling?
- How does DDIM choose which timesteps to use when running fewer steps?
- What's the trade-off between number of sampling steps and quality?
- Why doesn't DDPM produce good samples in 50 steps but DDIM does?

### Stretch Resources (Optional)

- **Implement DPM-Solver or DPM-Solver++:** https://arxiv.org/abs/2206.00927 — even faster than DDIM
- **Try image-to-image generation:** Take an existing image, partially noise it (don't go to $T$), then denoise. This is how SDEdit works.
- **Yang Song's blog on score-based models:** https://yang-song.net/blog/2021/score/ — unified theoretical view of diffusion
- Sanity check: verify that DDIM with $\eta = 1$ and 1000 steps produces samples statistically equivalent to DDPM

## Debugging & Reference

- **If DDIM samples are noticeably worse than DDPM at the same step count:** You probably have a sign error in the deterministic update equation
- **If timing isn't faster:** Verify you're actually skipping intermediate timesteps, not just iterating with smaller updates
- **Common mistake:** Indexing into the noise schedule arrays with the wrong timestep values when subsampling